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Provenance-Constrained Generation of Traceable Requirements for Model-Based Ship Systems Engineering

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ABSTRACT

Converting stakeholder requirements into traceable engineering specifications is a critical bottleneck in model-based systems engineering. Large language models can automate this conversion but produce requirement content—provisions, invoked standards, mandated actions, and numeric values—with no verifiable source, leaving engineers unable to tell which parts are grounded. We present a provenance-constrained generation workflow that prompts each "shall" statement to cite a knowledge base of *evidence units*—atomic, source-attributed statements extracted from regulatory and technical documents—organized by a 21-component × 45-aspect taxonomy, and to defer missing values explicitly. Applied to 60 ship cooling-system requirements, it attaches a citation to 98.6% of shalls, versus 0.0% for an unconstrained baseline, and yields zero uncited numeric commitments versus 69.2%. A random-retrieval ablation shows this follows from the citation-and-deferral constraint, while semantic retrieval raises the cited-value share (14.4% vs. 9.1%) and more than doubles regulatory citations. A post-hoc knowledge-base check assigns an any-value-supported label to 90.0% of selected RAG statements, while 92.2% of selected baseline statements have no supported or contradicted value in the retrieved evidence. These are statement-level labels, not full-statement correctness rates. A blinded LLM comparison gives mixed preferences, so the demonstrated contribution is source attribution rather than independently established accuracy.

SIGNIFICANCE AND PRACTITIONER POINTS

For researchers: We present a retrieval-augmented workflow that attaches source references to large language model (LLM)-generated engineering requirements: 98.6% of generated shalls carry a knowledge-base citation, versus 0.0% for an unconstrained baseline. A deterministic provenance-grade scheme (A/B/C/X) renders uncited numeric output measurable without a KB-truth-based correctness judgement; in the reported main RAG run the workflow produced 0.0% uncited numeric output (Grade X) versus 69.2% for the baseline. A random-retrieval ablation associates this outcome with the citation-and-deferral constraint (Grade X 0.0% in both retrieval variants), while taxonomy-aligned retrieval raises the share of regulatory cited values (21.6% versus 9.3%). The paper also reports where the workflow does *not* outperform the baseline: a knowledge-base check is limited by evidence coverage, and the blinded LLM comparison gives mixed preferences rather than establishing equivalence. The claim rests on source attribution, not independently established accuracy.

For practitioners: Engineers drafting requirements with LLMs cannot determine which parts of a generated requirement—an invoked standard, a mandated action, a constraint, or a value such as a pump startup time or alarm threshold—are grounded and which are fabricated. The proposed method prompts each shall to carry an explicit knowledge-base citation and missing values to be flagged as a structured outstanding item; applied to 60 ship cooling-system requirements it achieved 98.6% per-shall attribution (21.6% to regulatory sources) and produced no detected uncited numeric output. Outstanding values are delivered as structured placeholders naming the missing parameter,

expected unit, and source document—a concrete verification task rather than a confident but unverifiable specification—and a review route per shall directs engineer effort to the values whose evidence is weakest.

1 | Introduction

As ship engineering systems grow increasingly complex and integrated, tightly coupling mechanical, electrical, automation, and environmental subsystems, model-based systems engineering (MBSE) has emerged as the preferred framework for managing this complexity across maritime, aerospace, and defence industries [1]. Rather than relying on loosely linked documents, MBSE places the system model at the centre of the design record, making every engineering decision traceable through a structured hierarchy. Within this paradigm, engineering requirements occupy a foundational role: they are the primary bridge between stakeholder intent and formal system behaviour, and every downstream activity (architecture definition, component specification, verification planning, and classification submission) traces back to them [1, 2]. Engineering requirements produced in an MBSE workflow are expected to carry explicit source attribution: ISO/IEC/IEEE 29148 [3] requires each requirement to be traceable to an authoritative justification, and classification-society submission procedures enforce this linkage at the project level. For a ship main engine cooling water system subject to overlapping class rules from multiple classification societies, International Maritime Organization (IMO) conventions, and extensive manufacturer specifications, a complete set of such requirements may encompass several hundred individual shall statements, each requiring domain expertise to specify correctly and verify against the relevant regulatory provisions [4].

In practice, the conversion of raw stakeholder requirements into structured, traceable engineering specifications is the rate-limiting step of MBSE adoption for marine machinery systems (Fig. 1). This conversion requires a domain engineer to interpret each raw requirement against relevant regulatory documents, decompose it into individual engineering constraints, cross-reference corroborating or conflicting provisions across multiple sources, and express the result as a verifiable shall statement carrying an explicit source citation. For a single subsystem this typically occupies an experienced engineer for several weeks, and the resulting requirements often carry their regulatory basis as informal margin notes rather than as machine-readable citations, directly undermining the traceability chain that MBSE is designed to maintain [3].

Large language models offer a compelling route to automating this conversion: given a raw stakeholder requirement, an LLM can rapidly produce well-formed engineering shall statements that are syntactically correct and superficially plausible [5, 6]. The core problem is reliability. Every part of a generated requirement—the functional provision it states, the standard or classification rule it invokes, the installation, maintenance, or test action it mandates, the constraint it imposes, and any numeric criterion it specifies—emerges from statistical associations learned during pre-training rather than from a verified regulatory source [7]. The output may be syntactically correct and superficially authoritative while being factually wrong, outdated, or drawn from an inapplicable context, and it carries no link back to the provision that would justify it. Without explicit source attribution, neither the author nor a downstream reviewer can discern which parts of the requirement are grounded and which are fabricated. Numeric criteria provide a concrete instance: an uncited flow rate, pump-redundancy configuration, or pressure rating lacks an inspectable basis for downstream review. The provenance problem also extends to non-numeric requirement content.

Retrieval-augmented generation [8] makes grounding explicit by conditioning model output on passages retrieved from a curated corpus rather than on parametric recall, and has been shown to reduce factual errors in knowledge-intensive tasks [9]. Applying RAG to engineering requirement generation exposes limitations absent in open-domain question answering: knowledge-base entries carry categorical metadata (equipment class, functional intent, evidence strength tier) that semantic similarity scores alone cannot exploit; and the quality bar for generated requirements (traceability, verifiability, atomicity, non-ambiguity) is formally defined [3, 10, 11] rather than subjective, making it measurable.

Throughout this paper the retrievable unit of the knowledge base is an *evidence unit*: an atomic, source-attributed statement—a single regulatory provision, a design parameter, or a procedural requirement—segmented from an authoritative source document and carrying its document, clause or page locator, evidence tier, and taxonomy labels (Table 1). An evidence unit is therefore neither a whole document nor a fixed-length text chunk, and it is the object that each generated requirement cites.

An alternative to retrieval is to fine-tune a (small) language model on regulatory text, and small models fine-tuned on synthetic question–answer data and combined with RAG can rival larger models on the tasks studied by Soudani et al. [12]. We chose retrieval for this task for four reasons. First, parameter updates alone do not provide an inspectable clause-level citation. Retrieval supplies the justifying text and its identifier at generation time; a fine-tuned model could also be combined with such a citation mechanism. Second, classification rules can be revised, and a retrieval corpus can be updated and versioned without retraining or re-validating a model. Third, labelled pairs of stakeholder statements and regulation-grounded requirements, which supervised fine-tuning would need, are scarce in this domain. Fourth, controlled comparisons by Ovadia et al. [13] favour retrieval over unsupervised fine-tuning on their evaluated knowledge-intensive tasks. These findings do not establish a universal ranking of fine-tuning and RAG. A fine-tuned model would still be free to emit uncited numeric values, so the failure mode studied here would remain; combining a fine-tuned small model with the same citation-and-deferral constraint is a natural extension (Section 5).

This paper addresses these gaps through a knowledge-base–conditional, taxonomy-aligned RAG pipeline. Rather than letting the generator emit any plausible content, the pipeline conditions each shall statement on taxonomy-filtered retrieved evidence and constrains generation to attach KB citations and explicit deferral placeholders for missing values. The primary, directly measurable outcome is per-shall source attribution: 98.6% of generated shalls carry at least one KB citation, versus 0.0% for an unconstrained baseline. Within this cited-or-deferred output we additionally grade the dimension that is evaluated here—the numeric criterion—using a provenance-grade scheme: a numeric criterion is Grade A/B if it is backed by a citation, Grade C if it is an explicit deferral, and *Grade X* if it is a concrete value with no

verifiable citation—the untraceable "hallucinated" value that a downstream reviewer cannot distinguish from a correct one. The study makes no blanket claim that RAG output is of higher overall quality or more correct than unconstrained LLM output; the automated quality and KB-anchored checks do not establish independent correctness or equivalence, and we report their results and limitations in full. Its objective is narrower and verifiable: to attach source references to generated requirements and to drive the detected Grade X rate to zero. Crucially, because both per-shall citation and Grade X are defined by the presence or absence of a citation (a structural property) rather than by disagreement with the KB, the central finding does not depend on verifying KB-grounded output against the same KB and so avoids the circularity that affects KB-truth-based hallucination metrics. Fig. 2 provides a pipeline overview.

In summary, this paper makes three methodological contributions: a provenance-constrained generation workflow that attaches per-shall citations and explicit deferrals for missing values; a 21-component × 45-aspect engineering taxonomy that drives domain-aligned evidence retrieval; and a reproducible, non-circular provenance grading scheme (Grade A/B/C/X) for numeric criteria. The workflow is evaluated against an unconstrained LLM baseline and a random-retrieval ablation arm on 60 ship cooling-system requirements (Section 3). A requirement that carries its justifying citation at the moment of creation feeds directly into the downstream MBSE activities that depend on it—architecture allocation, component specification, verification planning, and classification submission—without the manual relinking step that informal margin-note citations force, and the resulting requirement set imports natively into a commercial MBSE tool (Section 3). In this sense the contribution is a generation step that yields requirements with explicit source-attribution fields, inserted at the rate-limiting boundary of the MBSE lifecycle, rather than a stand-alone text-generation result. The remainder of this paper is organised as follows. Section 1.1 reviews related work in requirements engineering, retrieval-augmented generation, traceability, and model-based tooling. Section 2 details the knowledge base, the taxonomy-aligned retrieval pipeline, provenance-constrained generation, and the experimental design. Section 3 reports the traceability and provenance-grade results together with correctness checks, a random-retrieval ablation, a side-by-side comparison, and a cross-generator robustness check. Section 4 interprets the findings and their limitations, and Section 5 concludes.

1.1 | Related Work

1.1.1 | Requirements Engineering for Complex Systems

Requirements engineering as a discipline addresses the systematic elicitation, analysis, specification, and management of system requirements [14, 15]. Requirements are organised hierarchically: stakeholder requirements state needs in the stakeholders' own vocabulary, system requirements restate them as verifiable technical obligations, and lower-level requirements are allocated to subsystems and components [1]. ISO/IEC/IEEE 29148 [3] prescribes characteristics of individual requirements (Section 5.2.5) separately from those of requirement sets (Section 5.2.6). Traceability is addressed through lifecycle relations (Section 6.4.3.5), rather than being a named individual-requirement characteristic. The conversion studied in this paper—from informal stakeholder statements to traceable engineering *shall* statements—is the stakeholder-to-system transformation at the top of this hierarchy. The formal foundations for safety-critical systems are laid out in ISO/IEC/IEEE 29148 [3] and the INCOSE Systems Engineering Handbook [1], which place traceability within requirements management and lifecycle practice. In the maritime sector, MBSE adoption has accelerated as classification societies shift toward digital rule delivery. The bulk of automated requirements engineering research has targeted quality assurance of existing specifications: detection of requirements smells [10] and NLP-based defect detection methods demonstrated at industrial scale [16, 17]. The upstream problem of generating new, evidence-grounded engineering requirements has received comparatively little attention. Zhao et al. [18] find that generation remains the least studied task, and domain-specific grounding is consistently the open challenge.

1.1.2 | Requirements Traceability and Model-Based Tooling

Beyond being a stated quality property, traceability is operationalised in practice as a maintained linkage between each requirement and the source that justifies it, and it is this linkage that classification-society submission procedures enforce at the project level. Gotel and Finkelstein [19] distinguish *pre-RS* traceability, concerning requirements production before inclusion in a specification, from *post-RS* traceability, concerning subsequent deployment. Either may be traversed forwards or backwards. The present work concerns source attribution during requirements production. Most automated traceability research is *recovery*-oriented: it reconstructs missing links between requirements and downstream artefacts after the fact, through information-retrieval and machine-learning techniques [19, 20]. Recent work applies LLMs and retrieval-augmented generation to the same recovery task: Hey et al. [21] evaluate RAG-based inter-requirement trace-link recovery on six benchmarks, with comparative performance varying by dataset and metric, and the LiSSA framework of Fuchß et al. [22] generalises RAG-based link recovery to requirements-to-code, documentation-to-code and documentation-to-architecture-model tasks. In systems engineering, Bonner et al. [23] combine embedding similarity with LLM classification to link requirements to MBSE model elements semi-automatically, and AI-assisted requirements traceability in MBSE for complex aerospace systems was presented at the 2025 SERC AI4SE & SE4AI workshop [24]. Such work presumes that the requirements already exist and that their source links were lost or never recorded—precisely the situation that informal margin-note citations produce. On the modelling side, SysML provides first-class constructs for requirements and their relationships [25]: «deriveReqt» points from a derived requirement to its source; «refine» from a refining element to the element it refines; «satisfy» from a design element to a requirement; and «verify» from a verification element to a requirement. A «trace» dependency expresses a general relationship. Nesting is containment, distinct from derivation, and a satisfy link alone does not demonstrate verified compliance; and commercial MBSE tools import these as navigable requirement diagrams; SysML-based architecture development has been carried through to full-ship

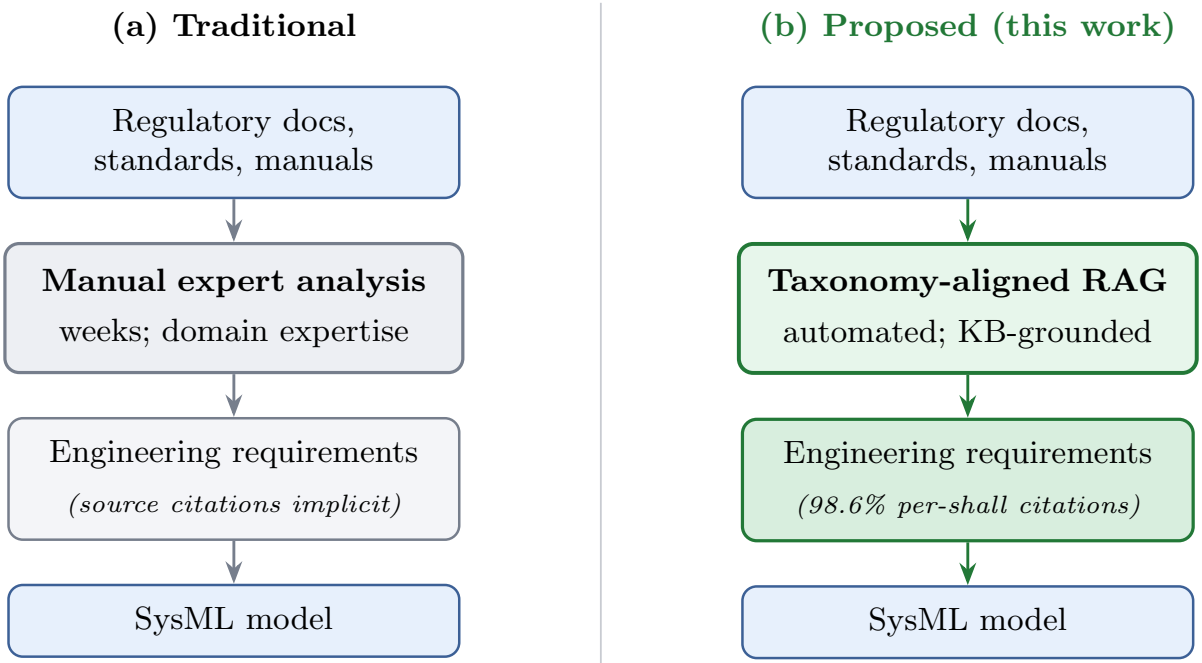


FIGURE 1 | Requirements engineering workflows compared. (a) Traditional: manual expert analysis is slow, expertise-intensive, and yields implicit source citations. (b) Proposed: taxonomy-aligned RAG pipeline automates the conversion and attaches explicit per-shall KB citations (darker box), supporting downstream traceability and classification-authority review.

subsystem integration in the naval domain [26]. Yet the tooling assumes that the requirements—and their citations—have already been authored by hand. Neither the recovery literature nor the modelling literature addresses the upstream act of *generating* requirements that are provided with machine-readable source citations at the moment of creation. The present work targets this gap directly: it attaches a per-shall citation (or an explicit deferral marker) during generation, and the resulting requirement set imports natively into a commercial MBSE tool without manual re-linking (Section 3).

1.1.3 | Retrieval-Augmented Generation

Lewis et al. [8] established that conditioning generation on retrieved passages substantially reduces factual errors in knowledge-intensive tasks; Izacard and Grave [27] further demonstrated that iterative passage retrieval improves generative accuracy across open-domain QA benchmarks. Hybrid retrieval combining sparse BM25 [28, 29] with dense vector search [30, 31] consistently outperforms either alone [32]—intuitively, the sparse component rewards exact matches of technical terms and identifiers (e.g. a component name or a clause number), while the dense component matches paraphrases that share no wording—and reciprocal rank fusion [33] provides score-agnostic fusion. For the dense component we use the BGE-M3 model (M3-Embedding [34]), built on BERT-style pre-trained transformer encoders [35], which handle mixed Chinese-English terminology naturally. Hierarchical RAG [36] and adaptive self-reflective RAG [37] share the motivation of enriching the retrieval context, but draw their structure from post-hoc clustering or reflection signals rather than from the inherent taxonomy of engineering documentation, and do not provide explicit metadata filtering. Attributed text generation—where every generated claim must cite a specific source—has been formalised as an evaluation task [38]; we apply analogous per-shall attribution requirements in the constrained setting of engineering specifications. Our contribution differs from this prior line in three respects: attribution is driven by a domain component–intent taxonomy rather than open-domain similarity; the target is safety-critical numeric commitments in engineering requirements rather than open-domain factual claims; and provenance is graded *deterministically* against the knowledge base—an uncited number is flagged regardless of how fluent or well-formed it appears, and cited identifiers are validated rather than taken on trust—instead of being scored by a model judge.

1.1.4 | Graph- and Ontology-Based Retrieval

A complementary line of work gives retrieval an explicit structure. GraphRAG (Edge et al. [39]) uses an LLM to extract an entity knowledge graph from the source corpus, pre-summarises communities of closely related entities, and answers

queries from these summaries, improving comprehensiveness on corpus-level questions; Peng et al. [40] survey graph-based retrieval-augmented generation, including knowledge-graph and ontology-guided variants in which typed entities and relations constrain what is retrieved. The 21-component × 45-aspect taxonomy used here plays the role of a lightweight ontology: rather than retrieving over an LLM-extracted graph, the pipeline filters retrieval to evidence units whose (component, aspect) labels match the requirement. We deliberately stop short of a full knowledge graph for two reasons. First, provenance requires each cited value to be quoted from an identifiable clause of a source document; graph extraction and community summarisation rewrite source text into model-generated entities and summaries, which would re-introduce exactly the unverifiable intermediate content the workflow is designed to exclude. Second, constructing and maintaining a domain graph over evolving classification rules is costly, whereas a fixed taxonomy over atomic evidence units already supplies the structural alignment that retrieval needs.

1.1.5 | LLMs for Requirements Engineering

Instruction-tuned large language models [41] can follow complex structured-output specifications, and chain-of-thought prompting [42] further enables step-by-step reasoning that is beneficial for decomposing stakeholder intents into atomic engineering constraints. However, faithfulness to source materials remains a persistent challenge: generation models produce outputs that are fluent and internally consistent yet factually inconsistent with their input context [7, 43]—the hallucination problem is fundamentally a faithfulness gap between parametric knowledge and authoritative sources. Ferrari and Spoletini [44] map the interface between formal requirements engineering methods and LLMs, identifying faithfulness, correctness, and traceability as the key concerns when deploying LLMs in requirements engineering workflows. Arora et al. [6] find that hallucination and lack of traceability are the primary quality concerns in LLM-generated requirements: outputs are syntactically well-formed but numerically unreliable when regulatory context is absent, and domain specificity is the dimension where models most frequently fail—precisely the failure mode that motivates the present work.

2 | Materials and Methods

2.1 | System Architecture

Fig. 2 provides an overview of the pipeline from stakeholder requirement to graded, attributable engineering requirements. The stages—requirement analysis, metadata-filtered hybrid retrieval, evidence value extraction, constrained structured generation, and provenance grading—are described in detail in Sections 2.2–2.4.

2.2 | Evidence Knowledge Base

Evidence units are the primary retrieval target, each covering a single design constraint, operational parameter, or regulatory provision. Every unit carries normalised metadata including a component category and an intent category (drawn from the 21×45 taxonomy), an evidence strength tier (L1 regulatory, L2 technical specification, L3 operational guidance), and source type.

TABLE 1 | An example evidence unit from the knowledge base (text translated from the Chinese original; all other fields verbatim).

Field	Value
Unit ID	`L1-CCS2025-7-251.6-001`
Text	"Cylinder cooling-water inlet pressure or flow alarm — Clause 7-25 1.6: when the cylinder cooling-water inlet pressure or flow is low, an alarm shall be given; when too low, slow-down and alarm shall be activated. For crosshead engines only the inlet-pressure item is required."
Source document	CCS *Rules for Classification of Sea-Going Steel Ships* (2025)
Locator	Clause 7-25 1.6, p. 1935
Evidence tier	L1 (regulatory)
Taxonomy labels	component = `cooling_system`; aspect = `alarm_setpoint`

The taxonomy was constructed by the authors from a functional decomposition of the cooling-water system: the 21 components group into rotating equipment, static equipment, piping, instrumentation and control, and working media, while the 45 intents enumerate recurring engineering aspects (e.g. parameter specification, redundancy, alarm and protection, installation, and maintenance). Each evidence unit is mapped to exactly one (component, intent) pair; of the 21×45 combinations, 183 are semantically valid. Tagging was LLM-assisted and author-reviewed. For units that already carried exactly one taxonomy-valid label from their source metadata, that label was inherited; for the remainder, DeepSeek was prompted with the unit text and the full taxonomy and asked to select the single best (component, intent) pair, and its choice was validated against the 183 valid pairs. Boundary cases that fitted no valid pair were confirmed one by one by the first author, and the remaining LLM-proposed labels were spot-checked by the authors; no systematic record of the spot-check was kept. Because this procedure involves a single LLM and no formal second annotation, an independent expert assessment has been prepared but no completed labels or agreement statistics are available (Section 4.6).

The KB contains 1,208 evidence units (L1 regulatory: 111, L2 technical: 455, L3 operational: 642) segmented from 38 distinct source-title entries—manufacturer project guides, engine manuals, installation guides, classification-society rules, IACS unified requirements, SOLAS, and US federal marine regulations—and covers 91 of the 183 valid (component, intent) pairs. It was assembled bottom-up: authoritative source documents were collected, segmented into atomic provisions, normalised, and tagged. The source count removes leading catalogue identifiers (such as S45) from source-title labels before deduplication: the same 1,208-unit audited file has 41 raw labels but 38 normalised titles. Three pairs of labels refer to the same CFR provisions with and without catalogue prefixes; this is a naming distinction, not a different KB version or a count of physical source files.

Provenance audit. Before the experiments reported here we audited every unit's provenance. An earlier version of the KB contained 108 units that had been drafted by an LLM to fill coverage gaps—gaps that had been identified by analysing the (component, intent) pairs required by the test requirements—together with a small number of consolidated summary units and units whose cited standard title did not match the actual standard. All such

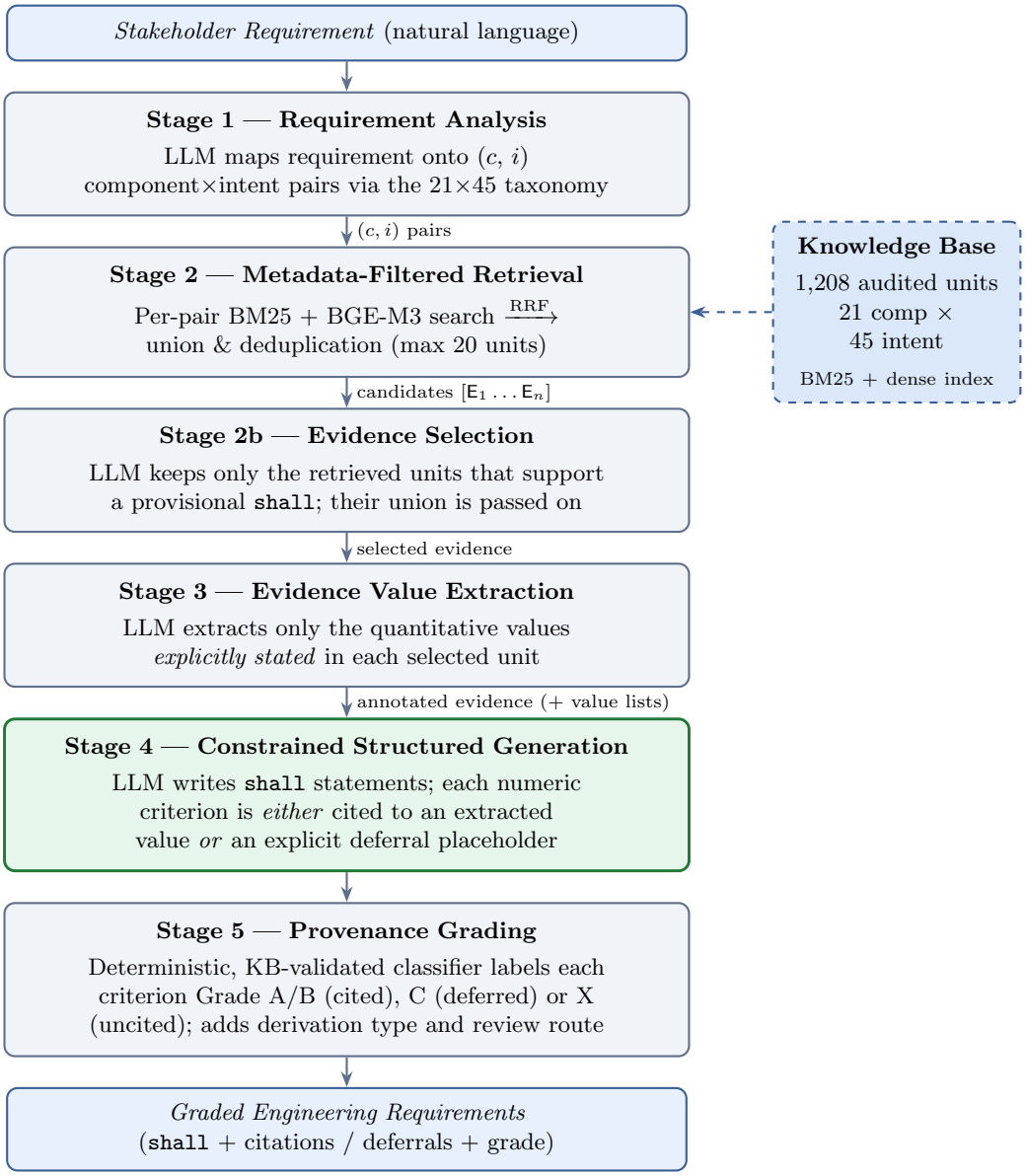


FIGURE 2 | Knowledge-base–conditional RAG pipeline. Stage 1 maps the source requirement to (c, i) component×intent pairs via the 21×45 taxonomy; Stage 2 restricts retrieval to KB units whose metadata matches those pairs and fuses results with RRF; Stage 2b keeps only the retrieved units that an LLM attributes to a provisional shall; Stage 3 extracts the quantitative values stated in the selected evidence; Stage 4 generates shall statements whose numeric criteria are either cited or explicitly deferred; Stage 5 assigns each shall a KB-validated provenance grade (A/B/C/X), a derivation type and a review route. The dashed arrow indicates KB access during retrieval.

units were removed; units whose content could not be located in an accessible original document were removed; and manufacturer and other non-regulatory documents previously tagged L1 were re-tiered to L2 or L3. Of the 1,364 pre-audit units, 156 were removed, 55 re-tiered and one source reference corrected; every result in this paper is computed on the audited KB. Because the KB remains a single-domain resource curated by the authors for the same cooling-water domain as the test requirements, the experiment should be interpreted as a controlled proof-of-concept rather than a fully held-out generalisation test (Section 4.6).

Evidence units are indexed with two parallel structures: a BM25Okapi index on tokenised text fields and an exact inner-product index on BGE-M3 unit-normalised vectors [34] (exhaustive brute-force search over the stored vector matrix, semantically equivalent to a flat inner-product index). Both indexes carry component-category and intent-category metadata fields used for retrieval filtering.

Table 2 summarises the audited KB by evidence tier and source type.

TABLE 2 | Composition of the audited knowledge base by source type and evidence tier (L1 regulatory, L2 technical specification, L3 operational guidance). Sources counts distinct source-title entries after removing catalogue prefixes, not physical files.

Source type	Sources	L1	L2	L3	Units
Classification rules, conventions, regulations	14	110	0	0	110
Standards	3	1	9	13	23
Manufacturer documents	21	0	446	629	1,075
Total	38	111	455	642	1,208

2.3 | Retrieval Pipeline

2.3.0.1 | Requirement Analysis. For each source requirement, a DeepSeek LLM [45] is prompted with the full 21×45 taxonomy and the requirement text. The model selects the (component, intent) pairs that are directly relevant to the requirement—with a recorded mean of 4.17 and range 2–9 pairs per requirement—and returns them as a JSON array. Pairs are validated against the taxonomy before use; if no valid pair is returned (e.g. for a broadly scoped requirement that does not match any single component), retrieval falls back to unfiltered search on the full KB.

2.3.0.2 | Metadata-Filtered Hybrid Retrieval. For each validated (component, intent) pair (c, i), two filtered searches are executed independently on the KB:

- *Filtered vector search*: cosine similarity is computed against the subset of KB units whose component category is c and intent category is i, returning the top-k units ($k = 8$ by default; varied in Appendix B). Similarities are read directly from the stored unit-normalised vector matrix for the matching subset; no index rebuild is required.
- *Filtered BM25 search*: BM25Okapi scores are computed over the same metadata-filtered subset, returning the top-k units.

The two ranked lists per pair are fused by reciprocal rank fusion (RRF):

$$\mathrm{RRF}(d) = \sum_{r \in \{\mathrm{BM25},\, \mathrm{vector}\}} \frac{1}{k_0 + \mathrm{rank}_r(d)}, \tag{1}$$

where $k_0 = 60$ is the standard smoothing constant [33]. Results from all pairs are merged by ID (deduplication), and the union is capped at $N_{\max} = 20$ evidence units. The candidate set therefore contains only units whose metadata aligns with the (component, intent) pairs extracted from the source requirement.

2.3.0.3 | Evidence Selection (Stage 2b). The candidate set is then narrowed by an LLM selection step: DeepSeek ($T = 0$) receives the source requirement and the numbered candidates, drafts provisional shall statements, and for each one lists only the candidate units that actually support it (each candidate is shown with its first 300 characters). The provisional statements are discarded; the union of the units they cite is passed to Stage 3. On the 60 source requirements this step retains on average 10.88 of 16.47 retrieved units.

2.4 | Engineering Requirement Generation

Generation proceeds in three stages (Stages 3–5 of the pipeline) that together enforce numeric traceability.

Stage 3 — evidence value extraction. Each selected evidence unit is passed to DeepSeek ($T = 0$) with a prompt that extracts only the quantitative values *explicitly stated* in the unit text (value, comparison operator, and unit), with an instruction never to infer or supplement a value. The extractor reads the first 1,200 characters of each unit. This produces, for every evidence unit, a list of grounded numeric parameters that the generator is permitted to use.

Stage 4 — constrained structured generation. The annotated evidence units are presented to the generator together with the source requirement. To bound prompt length, at most 12 units are shown, units with extracted values first, each with its first 500 characters of text followed by its full extracted-value list (the effect of these limits is measured in Section 3.8). The prompt requires the model to produce, for each shall statement, a measurable criterion that is *either* (i) a value taken from an evidence unit's extracted-value list, with the corresponding knowledge-base unit cited by identifier, *or* (ii) a structured deferral placeholder `{{TBD|unit|ref|component|intent|hint}}` naming the missing parameter, its expected physical unit, and the source document from which the value should be obtained. The prompt explicitly forbids writing any numeric value that does not appear in the supplied evidence (Appendix A), so that when the knowledge base lacks a precise figure the intended output is a structured deferral rather than a fabricated number. This is a prompting constraint rather than a hard decoding constraint; whether it held in practice is measured directly by the deterministic Grade X detector of Stage 5. Attribution is per shall rather than per requirement: a shall about pump start-up time cites only the units that state start-up-time requirements, not the heat-exchanger or piping units retrieved for other aspects of the same source requirement.

Stage 5 — provenance grading. Each generated shall is then classified by a *deterministic*, knowledge-base-validated classifier. A physical-unit regular expression scans the measurable criterion for a numeric value; each cited identifier is looked up in the KB, and identifiers that do not exist are discarded, so a model cannot obtain a grade by inventing an identifier; the evidence tier and the clause locator are read from the KB record rather than from the model's output. The classifier assigns Grade A (a numeric value citing at least one L1 unit whose KB record carries a clause, regulation, table or part locator), Grade B (a numeric value citing any valid unit), Grade C (an explicit deferral placeholder or a purely qualitative criterion), or Grade X (a numeric value with no valid citation). Because Grade X is decided solely by this regex-and-lookup rule, the Grade X rate is a fully reproducible measurement that invokes no KB-truth judgement and no LLM evaluation; it detects an uncited numeric criterion within its regex coverage, but cannot establish that a cited value is entailed by its source or detect every violation in free-text output.

2.4.0.1 | Derivation type and SysML relations. Each shall receives an additional deterministic *derivation-type heuristic*: *Unsupported* means no valid citation; *Deferred* marks a recognised deferral placeholder; *Direct* means the criterion's normalised numeric tokens occur in one cited unit; *Synthesized* means they occur across multiple cited units, or a qualitative criterion has multiple citations; and *Derived* is the remaining cited case. The

last label does not establish a valid calculation or a verified derivation chain. Token overlap likewise does not establish the parameter, operating condition or applicability. All labels therefore require engineering review. In the native EA export, source and child requirements are connected by *Nesting*; the derivation and review-route information is retained as properties. The conceptual «deriveReqt», «trace» and engineer-proposed «satisfy» links in Fig. 3 illustrate a traceability design, not automatically verified relationships in that export.

2.4.0.2 | Conflicting authoritative sources. The generator does not adjudicate between authorities. A deterministic screening rule flags candidate disagreements when cited units contain disjoint numeric tokens sharing the criterion's physical unit. This is not proof of a conflict: the parameter, equipment, operating conditions, applicable revisions and document authority may differ. Candidates are routed to a responsible engineer, who must establish applicability and resolve the governing flag-state, class, contractual and manufacturer requirements. Neither the latest date nor the stricter number is automatically selected. Absence of a flag is not evidence that all applicable documents agree.

2.4.0.3 | Review routing. Each shall also receives one of five review routes that tell downstream engineers what to do with it: *Adjudicate* (conflict flag), *Verify-or-reject* (Grade X), *Open item* (deferral), *Spot-check* (Grade A with a Direct derivation, i.e. numeric tokens matched to a clause-referenced regulatory unit), and *Engineer confirmation* (remaining cases, including qualitative uncited statements and requirements whose cited units are all non-regulatory, i.e. manufacturer or operational documents; we refer to such evidence as *tier-b* throughout).

The no-RAG baseline uses the same DeepSeek model and an identical structured-output instruction but receives no retrieved evidence and no extracted-value list; the model draws freely on parametric domain knowledge, and its citation list is empty by construction.

2.5 | Experimental Design

2.5.1 | Source Requirements

Sixty *source requirements* were constructed to span the ship main engine cooling-water domain across thirteen stakeholder roles covering the full system lifecycle, from design and procurement through construction, commissioning, classification approval, operation, and maintenance (Table 3). Most roles contribute four or five requirements, with a single cross-cutting project-manager requirement, so no single perspective dominates the set. Project documentation attributes their generation to GPT-5.5 (OpenAI), separately from the generation pipeline, using a prompt specifying the stakeholder roles and lifecycle phases of Table 3. The original interface/session record was not available in the inspected artifacts; the web-versus-API interface, account memory state and prior context therefore cannot be verified from the available artifacts. We do not claim independence on this basis and instead measured overlap between the inputs and the KB directly: the largest cosine similarity between any source requirement and any evidence unit is 0.816 (median 0.746), below the median nearest-neighbour similarity between two different evidence units (0.903); the longest character string shared with any unit has 11 characters; and 0 of the 60 inputs contain any numeral, which limits evidence of verbatim numeric copying but does not exclude semantic overlap or memory-mediated influence. The requirements are written deliberately as informal, underspecified natural-language statements—mirroring the way real stakeholders express intent (e.g. "the cooling system should stay reasonably stable and not let temperature, pressure, or flow swings disturb propulsion")—rather than as pre-formalised specifications; converting such statements into traceable, measurable shalls is precisely the task under study. They are synthetic but domain-informed and were not collected from a live shipbuilding project; accordingly the experiment is a controlled proof-of-concept rather than a field deployment (Section 4.6). Requirements were written in Chinese, reflecting the operational context of Chinese classification-society documentation and shipyard practice. The full set is held constant across all experimental arms.

Synthetic inputs are a deliberate proxy rather than a substitute for project data. They allow every lifecycle role to be represented in a controlled way and the set to be published in full, which project requirements protected by contracts usually cannot. They are, however, likely to be more grammatical and more uniformly scoped than requirements written by stakeholders, which often bundle several concerns, omit the operating condition, or use site-specific jargon. The pipeline handles such ambiguity in three ways that the measurements below make visible. A requirement that bundles several concerns maps to several (component, intent) pairs in Stage 1 (4.17 pairs per requirement on average, range 2–9); a requirement too vague to map to any pair falls back to unfiltered retrieval (0 of 60); and where the retrieved evidence does not determine a value, Stage 4 produces a deferral that names the missing parameter rather than resolving the ambiguity by guessing. Genuine human ambiguity is nevertheless a threat to external validity (Section 4.6).

2.5.2 | Experimental Arms

The main experiment compares two generation pipelines on the same 60 source requirements (Table 4). This comparison [46] holds the source requirements and generator endpoint constant, but changes both evidence conditioning and the citation-and-deferral prompt. It therefore compares workflows rather than isolating a retrieval-only causal effect. The random-retrieval ablation helps distinguish retrieval selection from the shared citation-and-deferral constraint.

2.5.3 | Random-Retrieval Ablation

To separate the contribution of the citation constraint (which enforces cited values or explicit deferrals regardless of evidence quality) from the contribution of semantic retrieval (which selects relevant evidence), we run a random-retrieval ablation arm on the same 60 source requirements. Rather than retrieving evidence by taxonomy-aligned hybrid search, this arm randomly samples—for each source requirement—the same number of evidence units from the full 1,208-unit KB (SEED=42, sampling without replacement) and passes them through the identical Stage 3 value extraction, Stage 4 constrained structured generation, and Stage 5 provenance grading pipeline. If the citation-and-deferral constraint is the operative mechanism, both retrieval

TABLE 3 | Design of the 60 stakeholder-style source requirements: thirteen roles spanning the system lifecycle. Counts sum to 60.

Stakeholder role	#	Typical concern	Lifecycle phase
Engine manufacturer	5	Parameters conform to engine technical data	Specification
System designer	5	Schematic completeness; circuit definition	Design
Equipment purchaser	5	Flow/pressure/material/interface verification	Procurement
Shipyard	5	Installation, welding, flushing, test access	Construction
Commissioning engineer	5	Pump pressure/flow and switchover verification	Commissioning
Classification surveyor	5	Compliance; low-supply alarm and switchover	Class approval
Ship owner	5	Operational stability and redundancy	Operation
Chief engineer	5	Centralised monitoring of T/p/level/pumps	Operation
Automation engineer	5	Sensor integration; trend/anomaly detection	Operation
Safety officer	5	High-temp/low-pressure/low-flow alarms	Operation
Maintenance supervisor	5	Cooler access, cleaning, inspection space	Maintenance
Quality inspector	4	Valve direction, connections, instrument fit	Inspection
Project manager	1	Consistency across spec, rule, and drawings	Coordination

TABLE 4 | Experimental arm definitions. Both arms use the DeepSeek API (`deepseek-chat`, recorded as `deepseek-flash` in the revision documentation) and the same 60 source requirements. ✓ = yes; × = no.

Arm	Retrieval	Evidence attribution	Description
RAG	✓	✓	Taxonomy-aligned hybrid retrieval; per-shall attribution
no-RAG	×	×	Strong baseline; unconstrained domain-knowledge use

variants should yield 0% detected Grade X; if semantic retrieval adds value over random selection, the Grade A/B share should differ between the two arms.

2.5.4 | Evaluation Metrics

The primary metric is the archived provenance-grade distribution; the remaining methods are auxiliary. Metrics excluded for structural bias are discussed in Section 2.5.5.

(i) Provenance-grade distribution (primary metric). Each generated shall is assigned a grade from the scheme of Section 2.4: **A** (precise value with an L1 regulatory citation and a clause reference), **B** (value backed by any cited evidence tier), **C** (explicit deferral placeholder), or **X** (concrete numeric value with no verifiable citation). The primary quantity is the Grade X rate, the fraction of shalls carrying an untraceable numeric value. Grade assignment is deterministic and does not invoke a KB-truth judgement: a shall is Grade X when its criterion matches the numeric detector and has no valid KB citation under the archived classifier. We additionally report the KB citation rate and the L1 (regulatory) citation rate, and decompose Grade C into deferrals where relevant evidence was retrieved but stated no precise value (an evidence-limited deferral) versus deferrals where the criterion is inherently qualitative.

(ii) Numeric correctness checks (whether emitted/uncited values are actually right). The grade scheme certifies citation, not correctness. A post-hoc knowledge-base assessment of selected numeric statements in both arms and a blind pairwise quality evaluation of the two arms are reported in Section 3.4. Both are computed on the same pipeline as the primary results.

An H_{hard} KB-anchored contradiction rate and a blind pairwise quality evaluation are reported in Section 3.4; as discussed in Section 4.3, neither is a discriminative measure of the traceability property this work targets.

2.5.4.1 | Supplementary Stage 1 mapping review.

For a limited mapping check, eight source requirements were selected before scoring from the remaining 58 after excluding the previously exposed OR001 and OR002 (fixed shuffle seed 20261017). Their four-method prediction union contains 75 candidate (component, intent) pairs. Two postgraduate reviewers with backgrounds in marine and systems engineering, who are fellow students of an author, reviewed separately and subsequently discussed their judgements. The authors confirmed that both reviewed the complete 183-pair taxonomy for each source and checked for omissions; this retrospective process confirmation is retained separately from the unchanged worksheets. No review-time measurement is claimed.

The saved DeepSeek mappings match the historical v3 Stage 1 records reused as upstream input by v4; the absence of a duplicated mapping field in v4 does not mean that this upstream mapping is untraceable. This does not equate the complete v3 and v4 experiments. Qwen here is a saved Stage 1 mapping baseline, distinct from downstream Qwen generation and the automated quality evaluator. BM25 and embedding predictions are also frozen; no new model calls were made for this supplement.

Uncertain and unlisted pairs were not converted to negatives. As a post-hoc supplementary analysis, all four methods are evaluated on the same mask of source–pair items for which both reviewers supplied definite binary labels after discussion. For this mask M, reference positives G and method predictions P, counts are $TP = |P \cap G \cap M|$, $FP = |(P \cap M) \setminus G|$ and $FN = |(G \cap M) \setminus P|$. Micro precision, recall and F_1 are conditional on M; mask coverage and excluded predictions are reported alongside them. Supplementary Material S1 provides individual-reviewer results, the fixed masks and uncertainty sensitivity, not a complete gold standard.

2.5.5 | Measurement Bias Analysis

Three metrics are excluded from comparative reporting due to structural biases. *Traceability rate* is 98.6% for RAG and 0.0% for no-RAG by construction, so it measures process compliance rather than output quality (though it enters the grade scheme as the necessary condition that defines Grade X). *Over-inference rate* structurally penalises any domain-knowledge extrapolation by no-RAG—which has no retrieved evidence—regardless of engineering soundness. H_{soft} is dominated by KB coverage rather than generation quality. The structural explanations are discussed further in Section 4.3.

2.5.6 | Implementation

The October 2026 project run records identify the DeepSeek API (`deepseek-chat`, reported in the revision records as `deepseek-flash`) for the main arms [45]; the cross-generator check uses `qwen-max` via the DashScope API. H_{hard} claim extraction uses DeepSeek and KB judgement uses Qwen. All LLM calls use temperature $T = 0$. Dense embeddings use BGE-M3 [34]; sparse retrieval uses BM25Okapi from the `rank-bm25` library; vector search is exact inner-product search over unit-normalised vectors. The project notes describe an Intel Core i9-14900HX laptop with an NVIDIA GeForce RTX 5060 Laptop GPU (8 GB); a contemporaneous hardware manifest was not found in the inspected artifacts. The evidence index is built once, offline, on the CPU, for the 1,208-unit index. The saved latency artifact records CUDA out-of-memory at model loading; project notes also report errors on long source segments. No successful GPU timing is available. At query time only one short requirement text is encoded per Stage 2 call (64.2 ms per query on the CPU), so a GPU would shorten only a negligible part of the run. Latency is dominated by the LLM API: the mean wall-clock time per source requirement was 0.93 s for Stage 1, 0.23 s for retrieval, 6.05 s for evidence selection, 9.13 s for value extraction and 9.67 s for generation (26.01 s in total, about 26.01 min for all 60 requirements run sequentially).

All reported analyses—the provenance-grade results (Section 3.2), post-hoc numeric verification and the correctness checks (Section 3.4), random-retrieval ablation (Section 3.3), and the leakage check—use the archived revision inputs and outputs. Stage 3 accuracy remains pending (Section 3.6). The output-grade analyses are computed on the single value-extraction and grading pipeline of Section 2.4 (513 semantic-RAG / 295 no-RAG shalls; 485 random-retrieval shalls for the ablation arm).

3 | Results

3.1 | Descriptive Statistics

Table 5 reports output volume and attribution statistics for both arms across 60 source requirements.

The two arms generate different output volumes (513 RAG vs. 295 no-RAG shall statements). The primary result is per-shall source attribution: **98.6% of RAG shalls carry at least one knowledge-base citation**, versus **0.0% by design for no-RAG**. This measures citation presence at the statement level; it does not establish that each provision, action, standard or value is semantically supported or applicable to the intended installation. The two arms also differ in how numeric criteria are expressed: 69.2% of no-RAG shalls state a concrete numeric value—all of them uncited—whereas RAG states a concrete, cited value in 14.4% of shalls and defers the remainder with an explicit placeholder rather than committing to an unsupported number. Numeric provenance is analysed in detail next.

3.2 | Numeric Provenance Grades

Within the cited-or-deferred output, Table 6 reports the provenance-grade distribution of numeric criteria.

The sharpest contrast is Grade X, the untraceable-value rate. The RAG pipeline produces **zero** Grade X shalls (0/513; Wilson 95% CI [0, 0.7%]): every numeric criterion it emits is either cited (Grade A/B, 14.4% of RAG shalls combined) or an explicit deferral (Grade C). The unconstrained baseline produces **69.2%** Grade X (69.2% of 295 shalls; 95% CI [63.7%, 74.1%])—that share of the baseline's shalls asserts a concrete numeric value (an alarm threshold, a flow rate, a switchover time) with no citation and no way for a downstream engineer to check it. Grade A (precise values carrying an L1 regulatory clause with a clause reference) is rare in the RAG arm (0.2%, 1 shall), because most L1 units in the knowledge base were extracted from multi-column rule tables without usable clause references; the broader measure of regulatory backing is the L1 citation rate, 21.6% for RAG versus 0.0% for the baseline. Because the deterministic Grade X detector (Section 2.4) registers every numeric criterion that lacks a citation, the zero-Grade-X result for RAG is a reproducible measurement rather than a KB-truth comparison; across all reported RAG runs the detector found no Grade X output.

Grade C is dominated by evidence-limited deferral, not evasion. Grade C accounts for 85.6% of RAG shalls; decomposing these 439 placeholders shows that 80.0% (351) are cases where relevant evidence was retrieved but stated no precise numeric value, so the pipeline correctly defers with a structured deferral marker recording the parameter and expected source; 18.5% (81) are inherently qualitative criteria with no numeric target; and only 7 are deferrals where no supporting evidence was retrieved at all. The high Grade C share is therefore largely a faithful signal of current KB coverage limits, not a failure to produce usable output: each deferral is an auditable instruction to the engineer rather than a fabricated value presented as a specification. The no-RAG baseline's 30.8% Grade C arises differently—these are qualitative shalls it phrased without a number—and it provides no mechanism to defer when it lacks a grounded value, which is precisely why its remaining numeric output falls into Grade X.

3.2.0.1 | Native import into a commercial MBSE tool.

An independent EA import/readback demonstration confirms that requirement elements, properties and Nesting links can be retained in a commercial MBSE tool. Two complementary views distinguish hierarchy from requirement content. Figure 4 shows the three-level R002 design-parameter hierarchy, including nested children below ER-R002-01. Figure 5 shows the complete English Notes of all four R001 children. Its two panels repeat the same folded R001 root for readability; they do not represent different source requirements or additional links. The exported node text and colours are retained, with nodes translated and the same Nesting connections rerouted for this two-panel layout.

TABLE 5 | Descriptive statistics for the RAG and no-RAG arms across 60 source requirements.

Metric	RAG	no-RAG
Total source requirements	60	60
Total engineering requirements	513	295
Shall statements containing a number	17.5%	11.9%
Per-shall KB attribution	98.6%	0.0%

TABLE 6 | Numeric provenance-grade distribution of generated "shall" statements (RAG n=513; no-RAG n=295). Each shall is graded by the traceability of its `measurable_criterion`: **A** – precise value with an L1 (regulatory) citation and a clause reference; **B** – value traceable to any cited KB evidence tier; **C** – explicit `{{TBD}}` deferral placeholder; **X** – a concrete numeric value with *no* verifiable citation. Grade X is the failure mode the cite-or-defer constraint is designed to prevent.

Grade	RAG	no-RAG
A (precise value + L1 clause) ↑	0.2% (1)	0.0% (0)
B (value + any citation) ↑	14.2% (73)	0.0% (0)
C (explicit TBD placeholder)	85.6% (439)	30.8% (91)
X (untraceable value) ↓	**0.0% (0)**	**69.2% (204)**
KB citation rate ↑	98.6%	0.0%
L1 (regulatory) citation rate ↑	21.6%	0.0%

Both views use an existing UI demonstration batch of 30 source requirements and 98 child requirements, separate from the 60-source, 513-output experiment. The demonstration retains identifiers, full requirement text, provenance grades, evidence references and review metadata as model properties. Its colours represent the demonstration's source/evidence-support categories, not the paper's A/B/C grades or verified correctness. These categories remain unverified engineering judgements; provenance grades are separate properties. Nesting indicates model containment, not independently validated derivation. Successful import and readback establish technical interchange for this sample, not semantic validity, requirement satisfaction or experimental performance on the 513-output population. The original vector exports, including the collapsed R001 and R004 installation views, are retained as accompanying figure files.

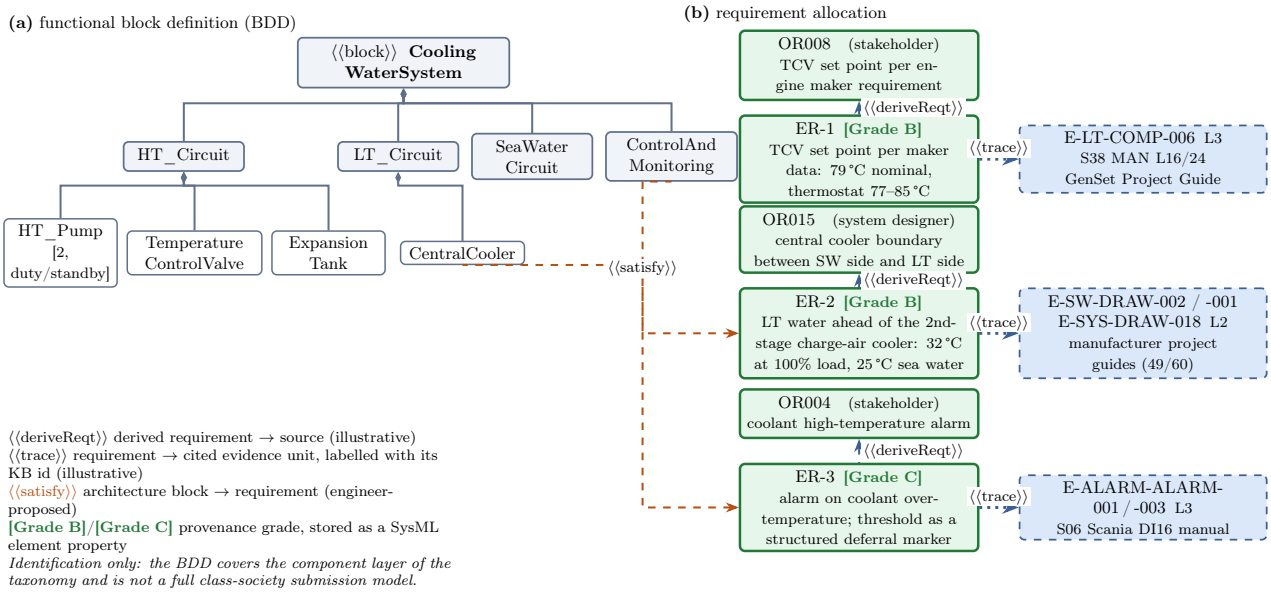


FIGURE 3 | Conceptual SysML context and traceability examples. **(a)** A block-definition view of the cooling-water component taxonomy; it is not a complete functional or class-society submission model. **(b)** Three requirement/evidence examples, with the second associated with OR015. Parameter examples come from different manufacturer/model contexts and do not constitute a jointly validated plant specification. Dashed «deriveReqt» arrows point from the child requirement to its source, and «trace» links point to the cited evidence units. Two engineer-proposed «satisfy» links point from blocks to requirements; satisfaction has not been verified. ER-1/ER-2 illustrate Grade B and ER-3 Grade C. These conceptual relations are not claimed as automatically verified native export links: the demonstrated native hierarchy uses Nesting and stores provenance and review information as properties.

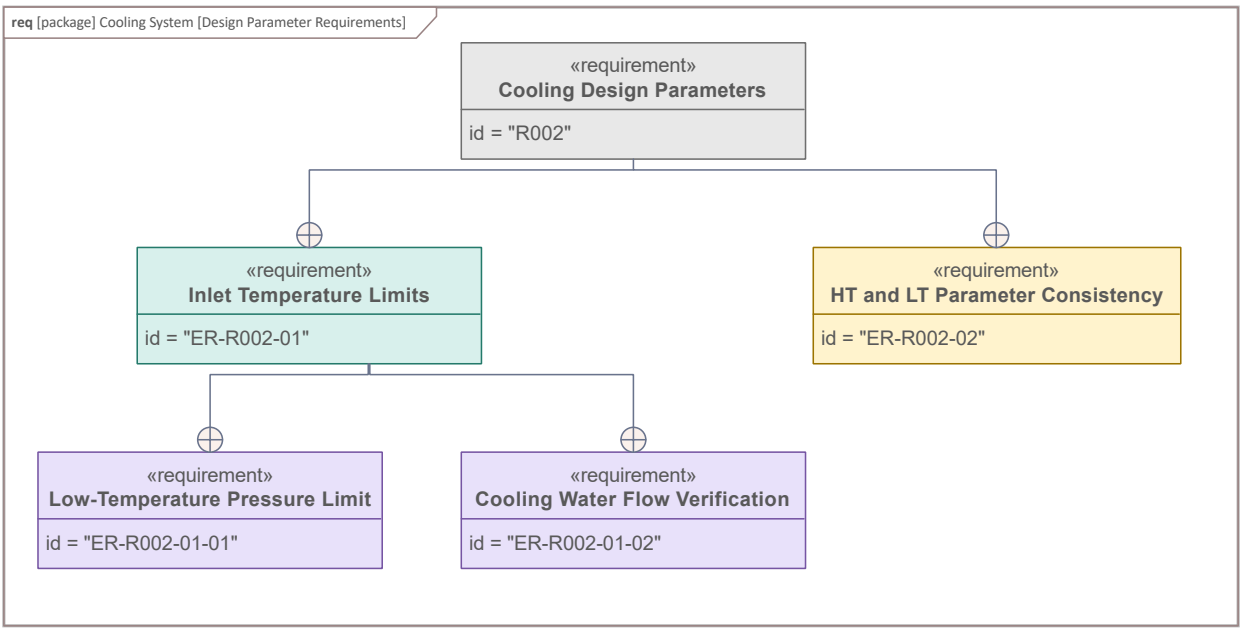
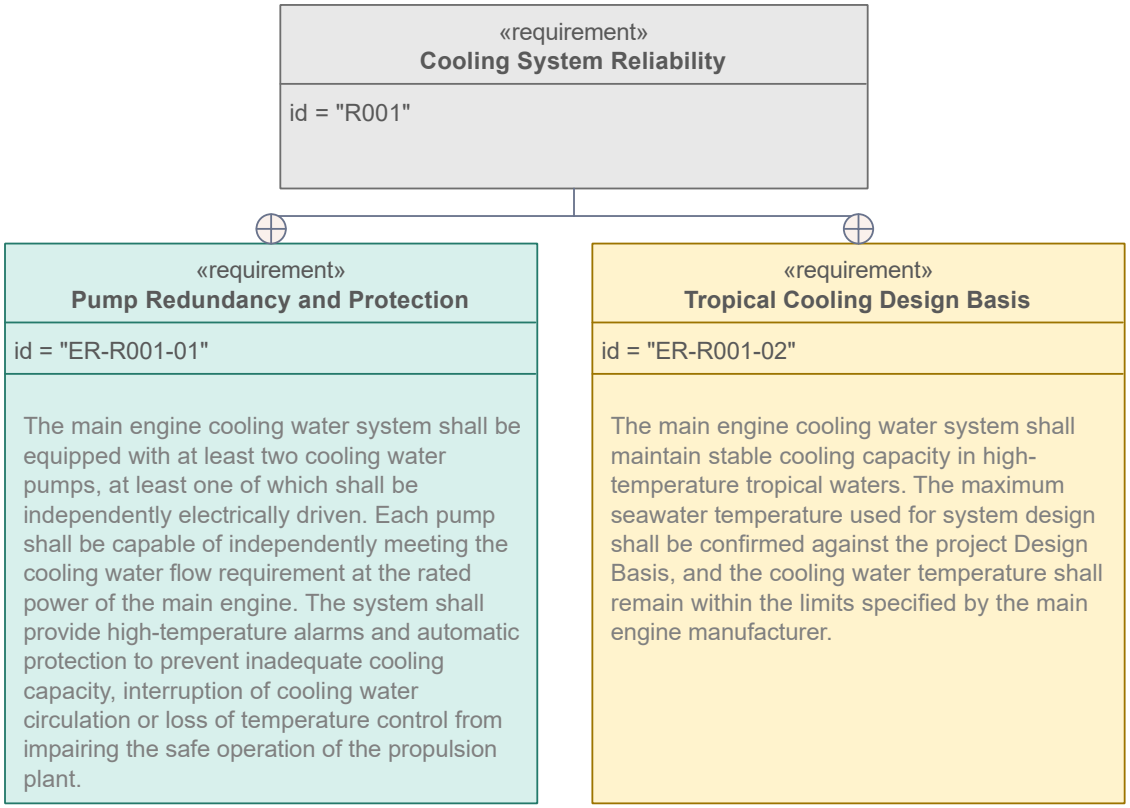


FIGURE 4 | Native EA design-parameter hierarchy (R002). The root, its children and the nested requirements below ER-R002-01 illustrate three levels of model containment. Nesting links do not establish validated derivation. This view belongs to the separate 30-source/98-child UI demonstration; colours are not correctness labels.

(a) R001 with children ER-R001-01 and ER-R001-02



(b) The same R001 with children ER-R001-03 and ER-R001-04

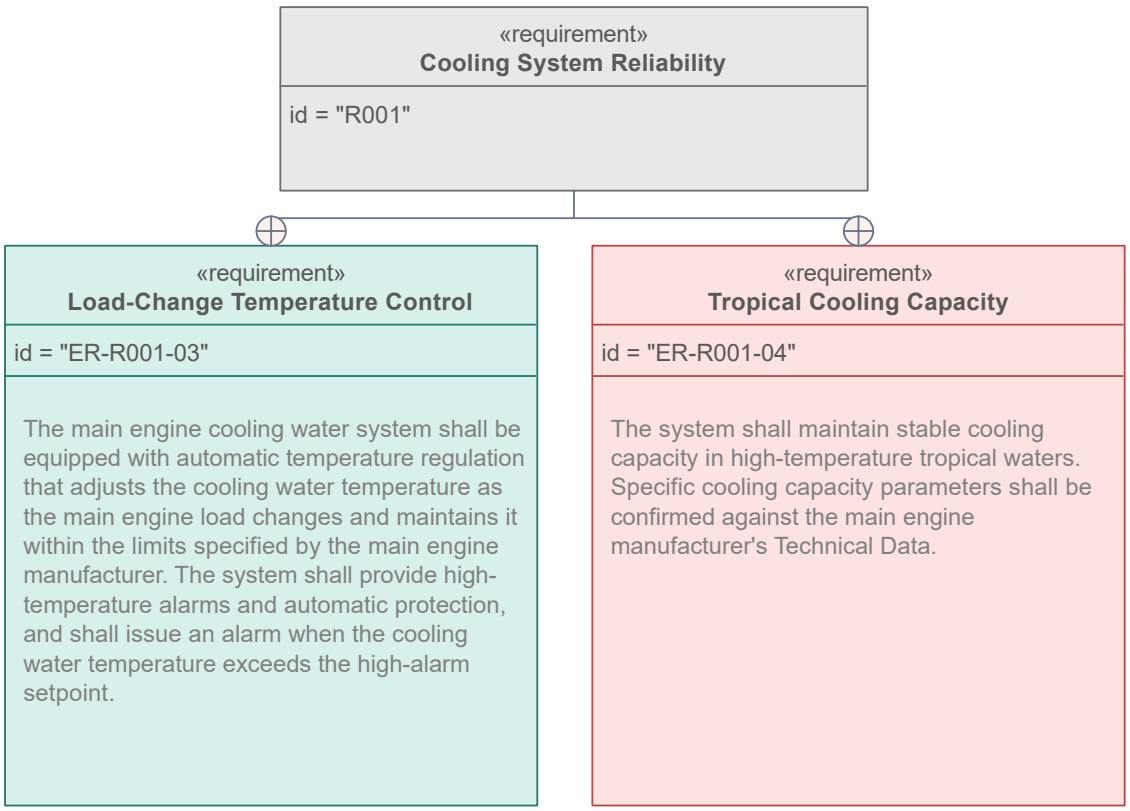


FIGURE 5 | Complete child Notes for the R001 reliability requirement. Panels (a) and (b) repeat the same folded root and partition its four actual Nesting links for readability. All child text and original colours are retained; only layout and link routing are adjusted. The view demonstrates retained requirement content in the separate UI batch, not validated engineering correctness or performance on the 513-output experiment.

3.3 | Retrieval Contribution

Table 7 reports the three-arm comparison across the 60 source requirements: taxonomy-aligned semantic RAG, random-retrieval ablation, and no-RAG baseline.

Two findings are notable. First, detected Grade X is 0.0% in *both* RAG variants, indicating that in the reported implementation the citation-and-deferral constraint—not semantic retrieval quality—is the mechanism primarily associated with the absence of untraceable values. Random evidence drawn uniformly from across all 1,208 KB units still drives detected Grade X to zero, because the generation prompt forbids uncited numeric output regardless of which evidence units are provided. Second, semantic retrieval improves the quality of the evidence that *is* cited: the Grade A/B share is 14.4% for semantic RAG versus 9.1% for random retrieval (about 1.6 times), and the L1 regulatory citation rate more than doubles (21.6% versus 9.3%). The random-retrieval arm also defers more (Grade C 90.9% versus 85.6%): when the retrieved evidence contains few grounded values, the generator defers rather than supplying cited numbers. The effect on the cited-value share is therefore smaller than the effect on regulatory backing, and the revised figures do not support a claim that semantic retrieval doubles the share of cited values; what it reliably improves is the proportion of requirements backed by regulatory rather than manufacturer or operational evidence.

3.4 | Correctness Checks on the Generative Content

Traceability is not correctness: a cited value can be consistent with its source and still be wrong for the application, and an uncited value can be accidentally right. Grade X measures the absence of a checkable link, not factual error, so we report three auxiliary checks: post-hoc KB assessment of selected numeric statements, an H_{hard} contradiction rate over extracted claims, and a blinded LLM pairwise quality evaluation. None provides independent human ground truth.

3.4.0.1 | Post-hoc KB verification of emitted values. The archived post-hoc check selects shall statements using a numeric-expression matcher, then asks DeepSeek ($T = 0$) to assess the extracted values against the cited units for RAG and the top-5 BM25-retrieved units for no-RAG. Its selected-statement denominators are 140 and 204, respectively; these are not the numbers of all claims or all criterion values in the two arms. Table 8 retains the archived aggregation: after the conflict check, a statement is labelled supported when at least one reported value is supported. Thus 90.0% and 4.9% are not all-values-supported rates or numeric-value precision.

For no-RAG, 92.2% of selected statements have no supported or contradicted value in the retrieved evidence; this is a retrieval- and KB-relative outcome, not proof that the values are unverifiable in general. The archived value-level records contain 291 supported verdicts out of 306 for RAG and 13 out of 489 for no-RAG. These use a different unit of analysis and do not establish complete sentence support, because extraction coverage and applicability have not been independently assessed. Two RAG rows have ambiguous matches to final output IDs and are retained as an audit limitation. The recorded conflict labels are one RAG statement and three no-RAG statements. One flagged RAG case concerns `E-HT-COMP-036` and a reported limit of −100 °C; its parameter and source context require review. The checks do not establish a general factual-error rate or an authoritative regulatory ruling.

3.4.0.2 | Paired conflict rate. Restricting to the source requirements for which both arms produced a numeric shall (n=34 pairs), the mean per-requirement conflict rate is 0.7% for RAG versus 2.0% for no-RAG (Wilcoxon signed-rank, p=0.56). This difference is not statistically significant at n=34; the paired-conflict comparison was significant on the pre-audit KB ($p = 0.0028$) but is not with the audited, cleaned KB used here, and we report the comparison as inconclusive rather than as evidence in either direction.

3.4.0.3 | H_{hard}: KB-anchored contradiction rate. This separate check extracts claims from `shall_statement` text using DeepSeek; it does not include the separate `measurable_criterion` field. For both arms, each claim is matched to the top-5 vector-retrieved KB passages and judged by Qwen as supported, contradicted or not found. The historical rate for each source requirement divides contradicted claims by *all extracted claims*, including not-found outcomes. It is not restricted to numeric or KB-checkable claims, and does not use RAG's own citation list as its retrieval procedure. Table 9 reports the unchanged mean of those source-level rates and the paired test over 60 source requirements.

The archived macro mean fractions are 0.0146 for RAG and 0.001 for no-RAG (Wilcoxon $p = 0.026$). The raw totals are 16 contradicted claims among 1,614 extracted RAG claims and 2 among 953 no-RAG claims; pooled rates from these totals are distinct from the reported macro means. The 915 and 273 supported-or-contradicted claims are only subsets, not the denominators used in the historical test. Moreover, historical API or parse failures were recorded as not found, and the retained records cannot

TABLE 7 | Retrieval ablation: semantic (hybrid) retrieval vs. random retrieval vs. no retrieval. Grade definitions as in Table 6.

Metric	RAG	Random	no-RAG
Total shalls	513	485	295
Grade A	0.2% (1)	0.4% (2)	0.0% (0)
Grade B	14.2% (73)	8.7% (42)	0.0% (0)
Grade C	85.6% (439)	90.9% (441)	30.8% (91)
Grade X	0.0% (0)	0.0% (0)	69.2% (204)
Grade A+B	14.4%	9.1%	0.0%
KB citation rate	98.6%	100.0%	0.0%
L1 citation rate	21.6%	9.3%	0.0%

TABLE 8 | Archived post-hoc labels for selected statements against the audited KB. Supported means at least one reported value is supported, not full-statement correctness.

Outcome	RAG	no-RAG
Selected statements	140	204
Any-value supported	90.0%	4.9%
Contradicted by KB	0.7%	1.5%
No supported/contradicted value	5.0%	92.2%
No numeric value returned	4.3% (6)	1.5% (3)

TABLE 9 | H_{hard} contradiction rate of all extracted claims (including not-found outcomes) per source requirement (paired Wilcoxon signed-rank test, $n = 60$).

Arm	Mean H_{hard}	Std
RAG	0.0146	0.0656
no-RAG	0.0010	0.0073
Wilcoxon $p = 0.026$ (RAG higher)		

separate these failures from absent evidence. This limits interpretation of the significant difference: it neither proves greater fabrication by RAG nor shows that the difference is caused solely by greater auditability. Applicability and independent correctness remain unresolved.

3.4.0.4 | **Blind pairwise quality evaluation.** Finally, we asked a separate LLM evaluator (Qwen) to compare the two arms' outputs side by side on four dimensions, with each pair presented in both left–right orders, without arm labels. A dimension receives a preference only when the two presentations agree; disagreement is counted as a tie. This reduces sensitivity to position but does not establish absence of evaluator bias.

The overall preference count is 64 for the RAG arm, 51 for no-RAG and 125 ties (out of 240 dimension-level comparisons), and the per-dimension counts include many ties: on *consistency*, the evaluator preferred no-RAG output 1 time and RAG output 0 times, preferring neither in 59 cases. An unblinded, single-dimension judgement on a pilot run had shown a large baseline advantage (107 to 26), but that judgement method is biased by knowing which arm produced which output; under the blinded protocol used here the counts give mixed preferences, with RAG ahead in the aggregate count. The 240 dimension-level outcomes are nested within 60 source requirements, not 240 independent expert judgements. No equivalence test was performed, so these counts do not establish equal quality or independent correctness.

3.4.0.5 | **Stage 1 mapping benchmark.** The eight-source review supplements the prepared 570-pair benchmark pool. In the submitted pre-discussion records, reviewer C labelled the 75 candidates as 25 relevant, 31 irrelevant and 19 uncertain; reviewer D labelled them as 34, 26 and 15, respectively. Label agreement was 59/75 (78.7%). After discussion, C's candidate labels were unchanged and D's counts became 26, 31 and 18. Agreement was then 74/75 (98.7%), including 18 jointly uncertain labels; this is post-discussion label agreement, not independent reliability or model accuracy. The remaining candidate disagreement concerns `flow_meter:alarm_setpoint` for OR032 (C uncertain, D relevant).

The reviewers listed four and twelve missing pairs, respectively, with a union of 13. Only one missing pair was explicitly relevant for both. Together with 56 jointly definite candidates, it gives a common 57-item mask (26 relevant, 31 irrelevant) within the 88 observed candidate-or-missing items: 57/88 (64.8%) coverage, retaining all eight sources. Table 11 reports conditional metrics on this mask. Its excluded predictions are not judged correct or incorrect, and absence from a reviewer's missing-pair list is not an explicit negative. These results do not establish complete mapping accuracy over the taxonomy or all 60 sources. Citation rates remain a different measure.

3.4.0.6 | **Status of the remaining human annotation instruments.** The S2 mapping supplement above is separate from the prepared 120-unit taxonomy-label review (S1), 50 individual blinded output judgements (S3: 25 RAG A/B and 25 no-RAG X), and 50-unit extraction review (S4). Completed results for S1, S3 and S4 remain unavailable; no taxonomy-label Cohen's κ or independent output-correctness score is claimed. S3 comprises individual judgements, not 50 matched pairs. One item matches the identical text of OR008.9 and OR008.10; that identity ambiguity precludes instance-specific attribution without further provenance evidence. These pending instruments are not evidence for the present claims.

TABLE 10 | Blind pairwise quality evaluation (RAG vs. no-RAG), Qwen evaluates both left–right orders; disagreement is counted as a tie. R = RAG wins, B = no-RAG wins, Tie = no preference or order-disagreement.

Dimension	R (RAG)	Tie	B (no-RAG)	Win rate (R)
Completeness	24	26	10	40.0%
Actionability	19	22	19	31.7%
Consistency	0	59	1	0.0%
Verifiability	21	18	21	35.0%
Overall	64	125	51	26.7%

TABLE 11 | Supplementary Stage 1 mapping: conditional micro metrics on the common definite-label mask (57/88 observed source–pair items, 64.8%; eight sources).

Mapping method	Total predictions	Included	Excluded	Precision	Recall	F_1
DeepSeek (historical Stage 1)	33	22	11	0.9545	0.8077	0.8750
Qwen (mapping baseline)	26	21	5	0.8571	0.6923	0.7660
BM25	32	26	6	0.2308	0.2308	0.2308
Embedding similarity	32	23	9	0.5217	0.4615	0.4898

The mask contains 56 jointly definite candidates and one jointly relevant missing pair (26 positive and 31 negative items). All methods use the same mask; all eight sources remain included. Predictions outside it are unscored, not treated as errors or successes. These are post-hoc restricted metrics, not complete-taxonomy or 60-source mapping accuracy. Supplementary Material S1 gives exact counts, source-level coverage and uncertainty sensitivity.

TABLE 12 | Blind correctness of numeric values (S3).

Pending expert blind assessment (workbook S3).

3.5 | Side-by-Side Comparison on Two Source Requirements

Table 13 gives the sentence-level comparison on two source requirements, one drawn from each regime, with the numbers unchanged from the result sets. Case 1 (OR008) illustrates attribution: both arms state temperature criteria, but only RAG cites the MAN L16/24 guidance. Applicability to the target engine still requires review. For Case 2 (OR010), the baseline's 30% lower glycol bound is below the 40–55% window recorded in the MTU A001065 KB unit `E-SYS-WATER-006` (L2). This is a comparison with a manufacturer-specific KB record, not proof that the baseline is wrong for every engine. The RAG arm did not retrieve that unit: it cites a Caterpillar freeze-point record and defers the concentration. The different manufacturer contexts must be checked before either output is used as a specification. The original manufacturer pages were not independently re-verified in this wording audit; the comparison demonstrates source attribution rather than validated technical suitability.

3.6 | Stage 3 Value Extraction Accuracy

Stage 3 extracts the quantitative values explicitly stated in retrieved evidence units; these values bound what the generator can cite without fabrication. The requirement phrasing of the extraction prompt is reproduced in Appendix A. The extractor was regenerated with the current generator and the current KB used here; earlier pipeline accuracy figures (precision 77.5%, recall 86.1%) no longer describe the reported run and are not quoted here. A re-measurement instrument has been prepared for a stratified sample of 50 evidence units drawn from the units the reported run actually used (across all three tiers), but independent expert values have not yet been supplied. Precision, recall and the per-tier breakdown therefore remain unavailable, as recorded in Table 14.

Two structural points hold independently of the pending measurement, and they are what matters for interpreting the grade distribution. First, the extraction step limits the Grade A/B share rather than the Grade X result: a missed value can lead to deferral under the prompt, so extraction completeness may shift outputs between Grade A/B and Grade C. This is intended behaviour, not a hard guarantee that a generator cannot emit an uncited value. Second, the known failure mode is concentrated in the L1 tabular sources: classification-society rules are laid out in multi-column tables, and the text-extraction step strips column–row relationships, so numeric values can be recovered without their parameter context (the flagged `E-HT-COMP-036` case at −100 °C illustrates why source context needs inspection; its precise failure mechanism has not been independently established). This lowers the achievable Grade A/B share for requirements that draw on tabular regulatory sources, and it is the reason the pipeline defers rather than guesses when a value's context is ambiguous.

3.7 | Cross-Generator Robustness

To test whether the workflow's behaviour depends on the specific generator, we repeated downstream value extraction and shall generation with `qwen-max` (Alibaba) replacing DeepSeek for value extraction and shall generation, reusing the identical retrieved evidence so that only the generator changes. Grades were recomputed by the same deterministic classifier, with cited identifiers extracted and validated against the knowledge base (Table 15).

The cite-or-defer effect transfers across generators. For `qwen-max`, retrieval-augmented generation reduces the uncited-numeric (Grade X) rate from 37.0% (no-RAG) to 0.0% (RAG), so both generator families reach a zero detected Grade X when retrieval

TABLE 13 | Side-by-side comparison of the two arms on the same source requirement. The Chinese source requirements and the generated shall statements are translated here; all numeric values are unchanged. Case 1 cites MAN L16/24 guidance; Case 2 compares the baseline with the MTU A001065 KB record (40–55%) while RAG cites a Caterpillar freeze-point record. These are manufacturer-specific contexts, not universally interchangeable limits. Original manufacturer pages were not independently re-verified in this wording audit. Attribution enables inspection but does not establish either output's applicability or correctness.

Src. req.	Arm	Shall (EN)	Criterion	Provenance (KB id, tier, document)	Grade & review route
OR008	RAG	The set point of the temperature control valve **shall** be verified against the engine maker's technical data; the auxiliary engine outlet temperature **shall** be held constant at approximately 80 °C.	≈ 80 °C	`E-LT-PRESS-003`, `E-LT-PRESS-005` (L2); S38 MAN L16/24 GenSet Project Guide, P255/P264	B; ENGINEER_CONFIRM (manufacturer guidance only)
OR008	no-RAG	The temperature control valve **shall** ensure that the cooling-water temperature is held within the range specified by the engine maker (stated as e.g. 70–85 °C).	Maker's range, stated as e.g. 70–85 °C	— (no citation)	X; VERIFY_OR_REJECT (numeric value with no traceable source)
OR010	RAG	The coolant freeze point and glycol ratio **shall** comply with the maker's data; pre-mixed ELC **shall** provide the specified freeze protection. (The concentration itself is carried as a deferral naming S26 MTU A001065.)	−37 °C	`E-COOLANT-WATER-065` (L3); S29 Caterpillar, *Know Your Cooling System*, P19	B; ENGINEER_CONFIRM (manufacturer guidance only)
OR010	no-RAG	The antifreeze ratio of the coolant **shall** comply with the coolant-system maker's concentration requirement, so that the freeze point and boiling point suit the operating environment.	Maker's concentration, stated as typically 30%–60% v/v; freeze point ≥ 10 °C below the local minimum	— (no citation)	X; VERIFY_OR_REJECT (numeric value with no traceable source)

TABLE 14 | Stage 3 extraction accuracy (S4).

Pending expert extraction review (workbook S4).

is constrained. The generator still matters for the other properties: `qwen-max` complies less reliably with the structured citation protocol, emitting citations as delimited strings rather than JSON objects and attaching a citation to only 91.7% of its shalls against 98.6% for the reported generator. Two implications follow. First, the benefit strengthens with the generator's instruction-following ability, so the zero-Grade-X result should be read as *achievable* under a sufficiently compliant generator rather than assured for every model. Second, grading must rely on *deterministic, knowledge-base-validated* citation checking rather than on model-emitted citation metadata: had `qwen-max`'s citation strings been trusted at face value, its traceability would have been substantially overstated.

3.8 | Sensitivity to Evidence Segmentation

The knowledge base is segmented into atomic evidence units rather than fixed text windows. Since each unit is the retrieval granularity and the citation target, segmentation artefacts would propagate into the grades, so we measured them directly on the 356 evidence units the reported run actually used. Of the 25 sentences in those units that contain both a condition and a numeric value, atomic-unit segmentation splits 0.0%, whereas fixed windows of 128, 256 and 512 characters with no overlap split 76.0%, 28.0% and 16.0% respectively (adding 20% overlap to the smallest window still splits 72.0%). These 25 selected conditional-numeric sentences form a boundary diagnostic, not a full end-to-end chunk-size experiment or a guarantee that all atomic units preserve context. The same measurement identifies truncation as a potential downstream information loss, without isolating its causal effect: since the evidence block shown to the generator truncates each unit at 500 characters, 65.4% of the used units exceed that limit and 24.1% of their numeric values lie past the truncation point. Those values remain available only through the separately extracted parameter list of Section 3.6, which is precisely why Stage 3 extraction accuracy bounds the achievable Grade A/B share, and why the pipeline defers rather than reads a value out of truncated context. The retrieval-variant and end-to-end runs (Appendix B) show where the outcome is and is not sensitive to these settings: the retrieved set barely moved with the BM25 fusion weight (Jaccard 0.947–0.971 against the default) or with the RRF smoothing constant k_0 changed from 10 to 100 (1.0), while reducing the per-pair k to four (0.64) or replacing the embedding model (0.788) changed it noticeably. Re-running the full pipeline for three end-to-end variants (vector-only retrieval, BM25-only retrieval and the mpnet embedding) left the grade outcome in the same range as the default arm: Grade X stayed at 0.0% in every variant and the Grade A/B share moved only within 10.9%–14.4%.

4 | Discussion

4.1 | Traceable Requirement Content as the Core Outcome

The central outcome is inspectable source attribution: cited shalls link to named knowledge-base units, and structured deferrals expose missing parameters. The archived grade detector found no Grade X output in the main RAG arm, compared with 69.2% in no-RAG. This is an observed result under the tested prompt and classifier, not a formal guarantee that the generator cannot emit an uncited number. The random-retrieval arm also reaches zero detected Grade X, separating this citation behaviour from the quality of retrieved evidence (Section 3.3).

Citation presence can be inspected without judging a statement against the KB, but a citation does not establish entailment, correct parameter context or applicability. Engineers must review cited as well as deferred content before committing a requirement to an architecture. The workflow makes that review more explicit; a reduction in review time or error has not been measured. The automated pairwise evaluation gives 64 RAG preferences, 51 no-RAG preferences and 125 ties, without establishing quality equivalence. Likewise, the 90.0% post-hoc supported label concerns selected statements with at least one supported reported value, not the proportion of all generated content that is correct. These boundaries preserve the distinction between attribution and validation.

TABLE 15 | Cross-generator comparison: grade distribution with DeepSeek versus `qwen-max` as the generator, on the same retrieved evidence. Cited identifiers are validated against the knowledge base before grading.

Generator	RAG Grade X↓	RAG A+B↑	no-RAG Grade X↓	no-RAG A+B↑
DeepSeek (`deepseek-flash`)	**0.0%**	14.4%	69.2%	0.0%
`qwen-max`	0.0%	40.2%	37.0%	0.0%

4.2 | Disentangling Citation Constraint from Retrieval Quality

The ablation (Table 7) separates two mechanisms. The citation-and-deferral constraint is the dominant factor behind the absence of detected Grade X: both retrieval variants reach zero, so for this requirement set and prompt the result did not depend on retrieval quality. Semantic retrieval instead governs *usefulness*—it raises the Grade A/B (cited-value) share from 9.1% to 14.4% and more than doubles the L1-citation share (9.3% to 21.6%)—because taxonomy-aligned retrieval preferentially surfaces classification-society rules whose structure mirrors the component-intent taxonomy, and the random-retrieval arm defers more often when grounded evidence is scarce (Grade C 90.9% versus 85.6%). The two effects are therefore separable, although the magnitude of the second is smaller than the constraint-and-retrieval pilot that motivated the design suggested: the constraint governs whether a numeric commitment is traceable at all, while retrieval quality governs how much of the output is backed by regulatory rather than manufacturer or deferred evidence.

4.3 | Why Conventional Evaluation Methods Fail to Detect the Traceability Advantage

Surface-quality judgements and citation checks measure different properties. A definite numeric statement may read well without identifying a source, while an explicit deferral may be less specific but easier to review. The blinded LLM comparison records mixed preferences (64 RAG, 51 no-RAG, 125 ties); it is not an independent correctness assessment or an equivalence test.

KB-anchored contradiction rates also depend on extraction and retrieval coverage. The historical H_{hard} divides contradicted claims by all extracted claims, including not-found outcomes, for each source requirement. Its paired macro mean is higher for RAG (1.5% versus 0.1%, $p = 0.026$). The 915 and 273 supported-or-contradicted claims are subsets, not those denominators. Historical failures are mixed with not-found records, and the check uses retrieved passages for both arms. Consequently, the difference cannot be attributed solely to citation visibility, nor does it establish a general fabrication rate. Neither auxiliary check demonstrates that constrained generation produces more correct content.

4.4 | Review Routing for Downstream Engineers

Provenance grades tell an engineer how a value is backed; the review routes of Section 2.4 tell the engineer what to do about it, and both are exported as element properties so that they can be filtered inside the MBSE tool. Table 16 describes the archived routing snapshot of 409 RAG shalls, which predates completion of the 513-output run; it is not a complete-run distribution. Of those 409, 73 were sent to engineer confirmation for manufacturer/operational-only evidence. Such evidence can be authoritative for a particular product but its applicability must be checked. The snapshot contains 317 open items, 91 engineer-confirmation cases and one spot-check. Its absence of candidate conflict flags does not establish agreement among all governing documents. The 204 verify-or-reject cases belong to the no-RAG arm, not to this RAG routing table. A later deterministic audit over all 513 outputs is a separate version and is not silently substituted into these archived results.

TABLE 16 | Derivation types and review routing for the archived, incomplete RAG routing snapshot ($n = 409$; not the 513-output result set).

Category	Count	% of shalls
Derivation type		
Deferred	317	77.5%
Direct	76	18.6%
Synthesized	4	1.0%
Derived	5	1.2%
Unsupported	7	1.7%
Review route		
Open item	317	77.5%
Engineer confirm	91	22.2%
Spot-check	1	0.2%

4.5 | Choice of Domain and Transfer to Other Ship Systems

The main-engine cooling-water system was chosen for three reasons. It is densely regulated by overlapping sources—IACS unified requirements, classification-society rules and manufacturers' project guides—so that the same parameter is often specified by several documents of different authority; its requirements are rich in numeric criteria (temperatures, pressures, flows, alarm and trip set-points, test pressures), which is where uncited values are most consequential and the provenance grades are most informative; and the authors had access to the full documentation set and to shipyard practice for this system. Other systems were not run in this study because building and auditing a KB is a substantial preparation effort in this project. The audit exposed important KB quality problems; it did not quantify costs or establish a universal cost ranking.

The retrieval, citation and grading workflow is potentially reusable, while the prompts, taxonomy and KB require domain

review. Performance outside cooling systems has not been established. Table 17 summarises how the 45 engineering aspects transfer. For the lubricating-oil and fuel-oil systems, which share the pump–cooler–tank–valve architecture, 38 of 45 aspects apply unchanged and the component list largely carries over with additions (purifiers, heaters, viscosity control); for HVAC and chilled-water systems 36 apply. For electrical power and ship networks, only 19 and 15 aspects apply unchanged, and we expect three further difficulties. First, their requirements are dominated by logical and timing conditions (selectivity, redundancy switch-over sequences, network segregation) rather than numeric thresholds, so the provenance grades would have to be extended from numeric values to logical conditions. Second, requirements at system interfaces (e.g. power supply to cooling pumps) need evidence from two knowledge bases at once. Third, electrical and automation rules rely heavily on tables, so the Stage 3 extraction problem reported for regulatory tables would be more acute. Replicating the study on the lubricating-oil system, the closest neighbour, is the first follow-up (Section 5).

TABLE 17 | Authors' assessment of how the 45 engineering aspects of the taxonomy transfer to other ship systems (Y: applies unchanged; M: applies after redefinition; N: not applicable). Per-aspect judgements are in the repository (`data/processed/taxonomy_transfer.csv`).

Target system	Y	M	N
Lubricating oil	38	3	4
Fuel oil	38	3	4
HVAC / chilled water	36	7	2
Electrical power	19	12	14
Ship network / automation	15	9	21

4.6 | Threats to Validity and Limitations

Internal validity. (i) Taxonomy labels were assigned with LLM assistance and author spot-checks; independent agreement has not yet been measured. A 120-unit stratified sample is prepared, but component-, intent- and pair-level Cohen's κ remain unavailable. Mislabelled units can be missed by metadata-filtered retrieval, which affects the cited-value share but not whether an emitted value is cited. (ii) The KB was curated by the authors for the same domain as the test set, and an earlier version contained LLM-drafted units targeted at gaps identified from the test requirements. These units were removed before the October revised runs on the audited KB. Subsequently, six empty RAG sets and five empty random-retrieval sets were regenerated at an 8,192-token cap using their saved evidence and extracted values; complete sets were retained. These phases should not be confused with a fresh rerun of every stage. Residual selection effects in which documents were collected cannot be excluded. Some retained regulatory units (from ABS, PRS and an industry standard) could not be re-verified against their originals during the audit because those documents were not available to us in full; they are flagged as such in the released audit sheet. (iii) The same LLM family performs generation and the auxiliary post-hoc KB verification; that auxiliary check is therefore not independent and is complemented by the blind pairwise evaluation. (iv) The supplementary mapping review uses a small, post-hoc definite-label mask: 57 of 88 observed items. Reviewer C had reassessed after feedback, so the submitted pre-discussion records are not claimed to be untouched blinded first passes. The reviewers are fellow students of an author; independence of initial work and full-taxonomy inspection are author-confirmed process accounts rather than timestamp-verified blinding. Excluding uncertain items can select easier cases, and candidate exposure can influence judgements. Missing-pair lists are not exhaustive explicit binary labels; the conditional metrics and supplementary sensitivity therefore do not establish full mapping recall.

Construct validity. The provenance grades measure whether a value is cited, not whether it is correct; correctness is addressed only by the knowledge-base-anchored checks and the blind pairwise evaluation of Section 3.4, both of which are limited by KB coverage and sample size. The supplementary Stage 1 check measures mapping judgements on a restricted mask, not downstream engineering correctness. A cited value can still be wrongly applied to a different component or operating condition; the review routes direct engineers to confirm such cases rather than assuming them correct.

External validity. All 60 source requirements address one system and were generated by an LLM rather than collected from a project; requirements written by stakeholders are likely to be messier (Section 2.5). Only two generator families were tested: both reached a zero detected Grade X with retrieval (`qwen-max`: 0.0% Grade X), but other properties such as the citation rate proved generator-dependent, so additional families remain to be tested. The DeepSeek endpoint is recorded differently in the initial and October revision documentation. The October main-arm runs and later selective refills are distinct phases (Section 2.5); an immutable provider snapshot was not found in the inspected artifacts, and absolute rates may change with model releases.

Conclusion validity. Proportions are reported with Wilson 95% intervals and paired comparisons with exact or rank tests. With 60 source requirements, power for modest differences is limited; the S1, S3 and S4 instruments still have no completed results. The eight-source mapping supplement is descriptive: its conditional metrics do not inherit the main experiment's 60-source inference, and its uncertainty bounds are not confidence intervals. No correction for multiple comparisons was applied to the secondary analyses. The reported grade tables retain the archived experiment version; later deterministic maintenance changes to placeholder and citation-ID interpretation are documented separately and must not be mixed with these totals.

Coverage and extraction. Only 91 of the 183 valid (component, aspect) pairs are covered by the audited KB, which caps the achievable cited-value share and drives the deferral rate; and the value extractor recovers only part of the values held in multi-column regulatory tables (Section 3.6), which lowers the achievable Grade A share. These limitations are distinct from citation presence; neither citation nor a deferral substitutes for checking source context.

5 | Conclusions

Generating engineering requirements from regulatory documentation for MBSE-based ship system design is a slow, expertise-intensive step that has resisted automation largely because LLMs produce requirement content—provisions, invoked standards,

mandated actions, constraints, and numeric criteria—that appears authoritative yet cannot be traced to a source. By aligning retrieval with a 21-component × 45-aspect taxonomy and prompting generation to cite supporting knowledge-base units or explicitly defer missing values, the proposed pipeline attached a citation to 98.6% of generated shalls, versus 0.0% for an unconstrained baseline. For the numeric dimension, the pipeline produced 0.0% detected Grade X outputs, versus 69.2% for the baseline.

The random-retrieval arm also has zero detected Grade X, supporting the role of the citation-and-deferral prompt for this sample. This observation does not prove a hard guarantee or establish that every cited statement is supported by its evidence. Most of the pipeline's Grade C output (80.0%) is evidence-limited deferral backed by retrieved-but-imprecise evidence, turning KB coverage limits into an explicit engineering worklist rather than fabrication. This property is invisible to conventional evaluation: the blinded LLM evaluation gave mixed preferences, and the KB-anchored contradiction rate is significant in the *opposite* direction, under a historical denominator that includes not-found and failed judgements (Section 3.4). The random-retrieval ablation attributes usefulness to semantic retrieval and the absence of untraceable values to the citation-and-deferral constraint. The workflow therefore claims auditability, not accuracy: it supplies inspectable citations and outstanding-item markers, while semantic support, applicability and correctness still require independent review.

Future work comprises two concrete tasks: (1) KB expansion to cover the remaining empty taxonomy pairs and to address L1 table-structure extraction (Section 3.6), which should shift Grade C mass toward Grade A/B while testing whether Grade X stays at zero; and (2) cross-domain replication on adjacent ship systems (fuel-oil, lube-oil, HVAC) to establish how the grade distribution generalises beyond cooling-water systems.

Author Contributions

Guangwu Liu: Conceptualization, Investigation, Writing — review & editing, Supervision, Resources, Project administration. **Dongsan Yan**: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing — original draft, Visualization. **Feihui Yuan**: Validation, Writing — review & editing. **Yuchen He**: Validation.

Conflicts of Interest

The authors declare no conflict of interest.

Data Availability

The project source repository is available at https://github.com/losyser/RAG_for_REQ. The existing archive identifier for the knowledge base, indexes and response caches is https://doi.org/10.5281/zenodo.20501537. A read-only check on 17 October 2026 resolved this concept DOI to version https://doi.org/10.5281/zenodo.20719799, dated 16 June 2026; the public GitHub main commit was `aab04d8`, also dated 16 June. These public releases predate the October revision and do not provide a verified release of its frozen experiment or human-review supplement. Public-release alignment remains outstanding before submission.

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Generative AI Statement

During the preparation of this work the authors used large language model tools (DeepSeek and Claude) to assist with language improvement and manuscript editing. The authors reviewed and edited all AI-assisted content and take full responsibility for the publication.

APPENDIX

A | Prompts

All prompts were issued in Chinese; below are English translations of the original Chinese prompts. Text in braces is filled in at run time; doubled braces in the source code are literal braces. The verbatim Chinese prompts are in the code repository (`cooling_rag_mg/requirement_analyzer.py`, `req_generation_v2.py`, `kb_value_extractor.py`, `structured_req_generator.py`).

A.1 | Stage 1: requirement analysis

DeepSeek, $T = 0$, max. 400 output tokens. System message:

```
You are a requirements-engineering expert for ship
    main-engine cooling-water systems.
Given a raw requirement, find all DIRECTLY relevant (
    component, engineering aspect) pairs
in the taxonomy below.
Rules:
- Select only pairs explicitly relevant to the
    requirement text; do not over-expand.
- Every pair must come from the taxonomy below (do not
     invent pairs).
- A requirement usually corresponds to 1-4 pairs;
    select more if it is broad.
- Output a JSON list; each item has the fields
    component and intent.
Taxonomy (component: [allowed aspects]):
  cooling_water_pump (cooling-water pump, incl. sea-
      water pump): [specification,
    redundancy, auto_start, material,
        corrosion_protection, installation,
        alarm_setpoint,
    trip_setpoint, pressure_test, inspection,
        maintenance, documentation,
    class_approval, spare_parts]
  cooler (cooler / heat exchanger): [specification,
      pressure_drop, material, ...]
  ... (21 components in total; full list in the
      repository)
```

User message:

```
Raw requirement: {requirement}
List all directly relevant (component, intent) pairs.
    Output only a JSON array, e.g.
```

```
[{"component": "expansion_tank", "intent": "
    level_control"},
 {"component": "expansion_tank", "intent": "
     alarm_setpoint"}]
```

A.2 | Stage 2b: evidence selection

DeepSeek, $T = 0$, max. 4,000 output tokens. The user message contains the raw requirement and the retrieved candidates numbered E1, E2, …, each shown as [*Ei*] (`component×intent`) followed by its first 300 characters. System message:

```
You are a senior ship main-engine cooling-water system
     engineer. Using the RETRIEVED
EVIDENCE, convert the raw requirement into structured
    engineering requirements (shall).
Requirements:
- Judge the relevance of each evidence item to the
    requirement and use only relevant
  evidence; give concrete engineering requirements and
       parameters according to the evidence.
- Do not invent specific parameters not covered by the
    evidence; you may write
  "to be confirmed against the manufacturer's
      technical data / rule clause".
- KEY: each shall cites only the evidence numbers (E1/
    E2...) that genuinely support that
  shall's content; if a shall has no supporting
      evidence, write evidence_refs as [].
Output JSON:
{"engineering_requirements":[
 {"shall_statement":"...","verification_method
     ":"...","acceptance_criteria":["..."],
  "evidence_refs":["E1","E3"]}, ...]}
```

Only the cited evidence identifiers are used downstream; the shall texts of this step are discarded.

A.3 | Stage 3: evidence value extraction

DeepSeek, $T = 0$, max. 600 output tokens, JSON mode; one call per evidence unit (first 1,200 characters).

```
You are an expert in extracting ship-engineering
    parameters.
Extract all explicit quantitative parameter values (
    number + unit) from the evidence text.
Evidence text:
{text}
Requirements:
1. Extract only numbers that explicitly appear in the
    text; never infer or supplement.
2. Each parameter has: value_text (original wording),
    operator (<=, >=, <, >, = ; empty
   string "" if there is no comparison operator), unit
       .
3. If there are no quantitative parameters, return an
    empty list.
Return JSON in the format {"values": [{"value_text":
    "...", "operator": "...", "unit": "..."}]}
```

A.4 | Stage 4: constrained structured generation (RAG arms)

DeepSeek (or `qwen-max`), $T = 0$, max. 4,000 output tokens, JSON mode. The evidence block lists at most 12 units (units with extracted values first), each as `[id] tier:{tier} source:{source document}`, its first 500 characters, and a line `[quantitative parameters] {extracted values}`.

```
You are a requirements engineer for ship main-engine
    cooling-water systems, generating
structured engineering shall requirements for MBSE/
    SysML modelling.
## Raw requirement
{source_requirement}
## Evidence list
{evidence_block}
## Output requirements
Return JSON: {"shalls": [<shall object>, ...]}
Each shall object contains:
- shall_statement: the shall statement (Chinese)
- measurable_criterion: the quantifiable criterion (
    see numeric rules)
- applicable_condition: applicable condition /
    operating mode
- verification_method: verification method (inspection
     / test / analysis)
- acceptance_criteria: list of acceptance criteria
- evidence_citations: list of cited evidence, each
    with kb_id / tier / source / has_clause
## Numeric rules (HIGHEST PRIORITY)
measurable_criterion may contain only one of two
    things:
1. a concrete value marked under [quantitative
    parameters] in the evidence list above,
   with the corresponding evidence cited in
       evidence_citations;
2. a placeholder {{TBD|unit:<unit>|ref:<source
    document>|component:<component>|
   intent:<aspect>|hint:<explanation>}}
It is strictly forbidden to write any number that does
     not appear in the evidence list.
All fields in Chinese.
```

A.5 | Stage 4: unconstrained generation (no-RAG arms)

Same model and settings; no evidence block.

```
You are a requirements engineer for ship main-engine
    cooling-water systems, generating
structured engineering shall requirements for MBSE/
    SysML modelling.
## Raw requirement
{source_requirement}
## Output requirements
Return JSON: {"shalls": [<shall object>, ...]}
Each shall object contains:
- shall_statement: the shall statement (Chinese)
- measurable_criterion: quantifiable criterion (fill
    in a concrete value from engineering
  knowledge, or write "cannot be determined")
- applicable_condition: applicable condition /
    operating mode
- verification_method: verification method
- acceptance_criteria: list of acceptance criteria
- evidence_citations: empty list []
All fields in Chinese.
```

B | Sensitivity Analysis

TABLE B1 | Retrieval-layer metrics across retrieval variants.

Variant	Units/req	L1 share	Numeric share	Jaccard
default	16.47	15.1%	31.6%	1.0
vector_only	15.43	17.7%	32.9%	0.757
bm25_only	15.43	14.6%	30.6%	0.746
w_bm25_0.25	16.47	15.4%	32.0%	0.971
w_bm25_0.75	16.47	14.4%	31.9%	0.947
k0_10	16.47	15.1%	31.6%	1.0
k0_100	16.47	15.1%	31.6%	1.0
k_4	13.7	16.1%	31.4%	0.64
k_12	16.93	14.3%	32.7%	0.805
embed_mpnet	16.55	16.3%	30.5%	0.788

TABLE B2 | End-to-end grade distribution across retrieval variants.

Variant	Shalls	Grade X	Grade A+B	Citation
default	513	0.0%	14.4%	98.6%
vector_only	487	0.0%	12.3%	100.0%
bm25_only	477	0.0%	10.9%	100.0%
embed_mpnet	497	0.0%	12.9%	100.0%

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